

DOCUMENT 00

Master Research Compendium

Project Knowledge Map · Requirements Traceability · TRL Dashboard · Research Gaps

The binding document for the Modular UAV Fleet Ecosystem project. Serves as a single-entry-point reference for navigating the full document set, tracking requirement coverage across documents, understanding the intellectual evolution of the project, and identifying open research questions. Does not duplicate content from DOC1–DOC8; cross-references only. A new team member reading this document plus DOC1 should be able to orient themselves fully within the project.

Project:	Modular Autonomous UAV Fleet Ecosystem
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1. DOCUMENT SET NAVIGATOR

The project is documented across nine files. DOC1 is the primary peer-reviewed paper. DOC2–DOC7 are supplementary technical reports covering IP strategy, advanced R&D;, carrier operations, failure engineering, manufacturing, and validation. DOC8 is the hardware specification compendium. This document (DOC0) is the master index.

Doc	Title	Core Question Answered	Read If You Need To Know...
DO C1	Synergistic Multi-Domain Optimisation of Autonomous UAV Fleet Systems	How do FOC, CFRP, AI scheduling, and fleet architecture interact to produce compound endurance gains?	The validated physics, measurements, and core architecture claims
DO C2	Patent & IP Strategy	What is patentable? Where does prior art block us?	Which inventions to file, how to differentiate from US11440679B2 and others, claim draft language
DO C3	Advanced Airframe Research	What aerodynamic and structural advances are planned for Phase 2–3?	Laminar fuselage CFD plan, rotor offset analysis, structural battery feasibility, ANC DSP design
DO C4	Carrier Operations & Logistics	How does the K1/K2 carrier platform actually work?	Battery swap procedure, energy budget analysis, multi-carrier relay protocol, K2 airborne rendezvous
DO C5	Failure Engineering	What can go wrong and how is it handled?	Complete FMEA, fault trees, reliability numbers, emergency procedures for every failure class
DO C6	Manufacturing & Commercialization	How do we build and sell this?	Full production BOM, PCB schematics, assembly procedure, cost-down to ■37,500/unit, market strategy
DO C7	Complete Validation Program	What tests are needed and what sample sizes are required?	L1–L5 test protocols, SIL/HIL environment spec, statistical gap analysis for all published claims
DO C8	Hardware Specification Compendium	What exactly is each drone unit, down to bolt level?	W1/W2/C1/K1/K2/D1/D2 unit specs, wiring diagrams, assembly guides, maintenance schedules

Table 1. Document set navigator.

2. PROJECT INTELLECTUAL EVOLUTION

The project passed through four distinct intellectual phases. Understanding this evolution prevents re-investigating dead ends and explains why certain concepts were abandoned.

Phase 0 — Per-Drone Optimisation (abandoned)

Initial framing: make a single drone more efficient through aerodynamic shaping (teardrop fuselage), FOC propulsion, and CFRP structure. This phase generated the list of innovations in the deep-research reports (structural batteries, morphing arms, push-flow propulsion, acoustic cancellation). It was abandoned when $W^{1.5}$ scaling analysis showed that adding capabilities to a single drone triggers a self-defeating power loop — every gram added increases required hover power at 1.5× the mass increase rate.

Phase 1 — Physics First (the pivot)

The $W^{1.5}$ analysis proved that the battery mass fraction optimality condition ($M_{b^*} = 2M_f$) requires a 1.65 kg drone to carry 1.10 kg of battery with only 0.55 kg of fixed hardware — impossible with any onboard AI compute. This mathematical result forced the architectural pivot: if compute cannot be on the worker drone, it must be elsewhere. This created the three-tier fleet model as a necessary consequence of physics, not as a product decision. Fleet architecture is the only engineering-rational response to $W^{1.5}$ at fixed drone mass class.

Phase 2 — Validated Subsystems

With architecture fixed, each subsystem was characterised: FOC propulsion (13 dB, 11.3× vibration), CFRP FEA (4.83 safety factor, 70% resonance margin), isolated power rail (eliminates 12% brownout), SPF navigation (92.7% indoor success), and the novel FOC→CFRP→EKF→SOC compounding chain linking propulsion electronics to battery state estimation accuracy (6.3% endurance improvement). These are the results in DOC1.

Phase 3 — Invention Synthesis (current)

With the core architecture validated at bench level, three operational gaps were identified: docking reliability in wind (alignment bandwidth problem → IR beacon), GPS-denied navigation without command-node dependency (SPOF problem → Federated EKF), and fleet scaling beyond 15 nodes (ESP-NOW limit → CBBA-SOC). These inventions are proposed and analytically grounded but not yet built. Phase 4 (Phase 1 hardware MVP) will validate or falsify them.

3. MASTER REQUIREMENTS TRACEABILITY MATRIX

Every requirement in the system specification is mapped to the document(s) that address it and the test(s) that verify it. Requirements with no test coverage are explicitly flagged as open.

Req. ID	Requirement	Addressed In	Verification Test	Status
MR-01	Fleet coverage ≥ 4 hr continuous	DOC1 §VI, DOC4 §2	DOC7 F7 (3 runs)	Open — carrier not built
MR-02	Waypoint success $\geq 95\%$ per mission	DOC1 §V-D	DOC7 S-01, F3	Partial — indoor 92.7%; outdoor TBD
ER-01	Worker hover endurance ≥ 120 min (W1)	DOC1 §VI-C	DOC7 F2 with W1	Open — W1 not built
ER-02	Worker hover endurance ≥ 38 min (W2)	DOC1 §V-A	DOC7 F2 — 38min measured	✓ Closed
DR-01	Docking success $\geq 99\%$ (calm)	DOC1 §V (mentioned), DOC8 D1	DOC7 F4 — 99% in 100 lab trials	✓ Provisionally closed (lab only)
DR-02	Docking success $\geq 95\%$ (8 m/s wind)	DOC3 §IR beacon concept	DOC7 F5 — 0 trials done	Open — critical gap
DR-03	Docking guidance bandwidth ≥ 100 Hz	DOC3 §5 (IR beacon), DOC8 D1	DOC7 IR-02	Open — IRDG-01 PCB not built
RR-01	Brownout rate = 0% at full AI load	DOC1 §V-E	DOC7 H-03 — 0% in 30 flights	✓ Closed
RR-04	IP54 minimum (outdoor components)	DOC8 all units	DOC7 — no IP test defined yet	Open — IP test protocol needed

Req. ID	Requirement	Addressed In	Verification Test	Status
NR-02	Position accuracy ≤ 0.30 m (cooperative)	DOC1 §III, DOC3 Federated EKF	DOC7 S-04 — SIL only	Open — SIL needed; flight TBD
NR-03	Federated EKF ≤ 0.20 m at 60 s	DOC3 §Federated EKF	DOC7 S-04	Open — plugin not built yet
CR-01	Command latency ≤ 200 ms	DOC1 §III-B	DOC7 L1 link test	Partial — LoRa only; dual-link not tested
CR-02	Mesh latency ≤ 50 ms (10 nodes)	DOC1 §III-B	DOC7 S-06 — 20 ms at 10 nodes	✓ Closed for ≤ 10 nodes
SR-03	Max fleet 15 nodes (ESP-NOW)	DOC1 §III-B	DOC7 S-06 at 15 nodes	Open — tested to 10 only

Table 2. Requirements traceability matrix. Open items are blockers for operational demonstration.

4. TECHNOLOGY READINESS LEVEL DASHBOARD

TRL scale used: 1=Basic principles; 2=Concept formulated; 3=Analytical proof; 4=Lab validation; 5=Relevant environment validation; 6=Demo in relevant env; 7=Prototype demo in operational env; 8=System complete; 9=Proven in operations.

System / Subsystem	Current TRL	Evidence	Next TRL Milestone	Est. Time
W ^{1.5} hover power model	9	DOC1: $<0.1\%$ error, $n=50$	Already at 9	—
FOC propulsion (T-MOTOR F90+F55A)	7	DOC1: anechoic, bench, flight	8 — validate on final W1 hardware	2 months
CFRP frame (Tarot TL65B44)	7	DOC1: FEA, flight, vibration	8 — fatigue life testing	6 months
Isolated power rail (PM02D)	8	DOC1: 0% brownout in 30+ flights	9 — 200-flight reliability	4 months
Energy-aware AI scheduling	7	DOC1: 11.4% gain in 10 flights	8 — validate on W1 with different payload	2 months
SPF navigation (indoor)	6	DOC1: 92.7% in purpose-built course	7 — outdoor varied environment	6 months
Standard GPS navigation (PX4)	9	Mature open-source stack	Already at 9	—
Autonomous docking (calm, lab)	5	DOC8: 99% in 100 lab trials	6 — outdoor calm conditions	3 months
Autonomous docking (wind)	3	DOC3: analytical bandwidth analysis	5 — outdoor trials 4/8 m/s	6 months
IR beacon docking guidance	3	DOC3: analytical design, latency budget	4 — IRDG-01/DOCB-01 bench test	4 months
Fleet CBBA-SOC scheduling	3	DOC1: algorithm design; DOC7: SIL spec	4 — ROS2 simulation	5 months
Federated EKF GPS-denied nav	3	DOC3: analytical derivation	4 — ROS2+Gazebo SIL	5 months
K1 ground carrier	3	DOC4: full spec, energy analysis	4 — K1 fabrication and static dock test	4 months
K2 airborne carrier	2	DOC4: specification only	4 — K2 build and hover test	12 months
Laminar teardrop fuselage	2	DOC3: CFD plan, geometry spec	3 — ANSYS CFD analysis	3 months
Rotor vertical offset (z/D)	3	DOC3: DLR data cited, geometry specified	4 — thrust stand comparison	1 month

System / Subsystem	Current TRL	Evidence	Next TRL Milestone	Est. Time
Structural battery arms	2	DOC3: feasibility analysis	3 — single-arm bench prototype	12 months

Table 3. Technology Readiness Level dashboard across all subsystems.

5. OPEN RESEARCH QUESTIONS

These are questions where the project does not yet have an answer. They are not gaps in documentation — they are genuine scientific unknowns that require experimental investigation.

Q1 — Docking success at high wind

- What is the actual docking success rate vs wind speed relationship for the IR beacon system? The analytical prediction suggests $\geq 95\%$ at 8 m/s and $\geq 80\%$ at 12 m/s, but this depends on the IR beacon control bandwidth being sufficient to reject wind disturbances during final approach. This is the single most important unknown for outdoor operational deployment.
- **Required:** F5 flight test campaign (DOC7). 30 trials per wind condition: 0, 4, 8, 12 m/s. If success $< 80\%$ at 8 m/s, the docking speed limit constraint must be revised.

Q2 — Federated EKF convergence in real hardware

- The Federated EKF has been analytically designed and the expected $\sigma_{\text{pos}} \leq 0.20$ m at 60 s derives from sensor noise assumptions ($\sigma_{\text{UWB}} = 0.10$ m, $\sigma_{\text{bias}} = 8$ μg). In practice, UWB multipath in buildings and near vehicles degrades ranging accuracy. The real-world positioning accuracy under field conditions is unknown.
- **Required:** SIL test S-04 first (validate algorithm); then outdoor HIL; then F-series flight. Accept criterion: $\sigma_{\text{pos}} \leq 0.30$ m at 60 s (relaxed from analytical 0.20 m to account for multipath).

Q3 — FOC compounding chain outdoors

- The FOC→CFRP→EKF→SOC chain was measured in a near-static indoor hover environment. Whether the 6.3% endurance extension survives in turbulent outdoor conditions — where accelerometer noise is dominated by wind-induced body vibration rather than motor vibration — has not been measured. Outdoor wind may wash out the CFRP damping benefit.
- **Required:** Repeat SOC estimation accuracy measurement during outdoor F3 flights at 0, 4, and 8 m/s wind; compare EKF SOC error vs indoor baseline.

Q4 — ESP-NOW mesh performance at 15 nodes

- Mesh latency has been measured at 10 nodes (20 ms). The 15-node hard limit for the protocol is theoretical; actual contention behaviour — packet collision rate, effective throughput per node, TDMA slot utilisation — at 13–18 nodes under full mission load (CBBA-SOC messages + telemetry + position injection) has not been measured.
- **Required:** SIL test S-06 with full message load at 10, 13, 15, 18 nodes. Document latency vs node count. Identify the practical operational ceiling below the theoretical 20-node limit.

Q5 — Carrier drone self-endurance under full recharge load

- The K2 carrier hover endurance under 2-bay recharge load ($360\text{ Wh} \div 416\text{ W} = 52\text{ min}$) is a first-principles estimate. Real K2 hover power depends on actual frame mass, actual battery discharge curve, and ambient conditions. Whether 52 min is achievable or whether the real figure is 35–40 min determines the K2 sortie planning entirely.
- **Required:** K2 build and full-load hover endurance test (3 runs). If $< 40\text{ min}$: either increase K2 battery capacity or reduce recharge load to one bay at a time.

6. CRITICAL PATH TO OPERATIONAL DEMONSTRATION

The following sequence is the minimum set of activities required to demonstrate 7+ hour continuous coverage in the field. Items are ordered by dependency: each requires the previous to be complete. Items marked BLOCKING cannot be bypassed.

Step	Activity	Blocker For	Est. Duration	Owner
1	Build W1 stripped worker drone (Sec DOC8-W1)	All W1 endurance claims	3 weeks	Hardware team
2	Validate W1 hover endurance (target 159 min)	Fleet endurance model	1 week	Test pilot
3	Build and bench-test IR beacon system (IRDG-01 + DOCB-01)	Outdoor docking claims	4 weeks	Electronics team
4	Fabricate K1 ground carrier with 2× D2 docking bays	All fleet rotation claims	5 weeks	Mechanical team
5	Static docking bench test (DOC7 F4): 50 trials calm	Wind docking tests	1 week	Test team
6	Outdoor docking wind test F5: 60 trials at 4 and 8 m/s wind	DR-02 requirement closure	2 weeks	Test team
7	Run SIL tests S-01 to S-05 (federated EKF, CBBA-SOC)	Fleet coordination claims	6 weeks	Software team
8	Implement CBBA-SOC on actual hardware (2 workers + K1)	Fleet rotation demonstration	4 weeks	Software team
9	Flight test F6: 2-worker rotation cycle (20 cycles)	Full fleet test	2 weeks	All teams
10	Flight test F7: full 2-hour fleet demonstration (3 runs)	Primary project milestone	3 weeks	All teams
11	Analyse results; close open requirements; update DOC1	Journal submission	3 weeks	Lead researcher

Table 4. Critical path to operational demonstration. Steps 1–6 must all complete before Step 7 begins.

7. COMPLETE MASTER BIBLIOGRAPHY

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8. TECHNICAL GLOSSARY

Term	Full Form / Definition
ANC	Active Noise Cancellation — real-time generation of anti-phase acoustic signal to cancel noise
CBBA	Consensus-Based Bundle Algorithm — decentralised multi-agent task assignment algorithm
CBBA-SOC	CBBA extended with State-of-Charge weight function for drone battery rotation
CFRP	Carbon Fibre Reinforced Polymer — T700/Epoxy composite used for drone arms and frame
CG	Centre of Gravity — must be within ± 5 mm of motor plane centroid for stable flight
DAF	Dynamic Amplification Factor — ratio of dynamic response to static response near resonance
DL	Disc Loading — total thrust divided by total rotor disc area (N/m^2)
DOD	Depth of Discharge — fraction of battery capacity used (80% DOD = 20% SOC remaining)
ELRS	ExpressLRS — open-source low-latency RC link protocol, <1 ms at 500 Hz
EKF	Extended Kalman Filter — nonlinear state estimator used for SOC and position estimation
FM	Figure of Merit — ratio of ideal induced power to actual shaft power ($0 < \text{FM} \leq 1$)
FOC	Field Oriented Control — sinusoidal motor commutation eliminating torque ripple
FTA	Fault Tree Analysis — top-down deductive failure analysis using Boolean logic trees
FMEA	Failure Mode and Effects Analysis — bottom-up systematic failure identification
HIL	Hardware-in-the-Loop — test methodology connecting real hardware to simulated environment
IR	Infrared — 940 nm wavelength used in the docking beacon guidance system
IRDG-01	IR Docking Guidance PCB — custom drone-side photodetector and demodulator board
$k_{\text{empirical}}$	Empirical hover power constant: $0.639 \text{ W/N}^{3/2}$, calibrated from W_1/W_2 measurements
LoRa	Long Range — 900 MHz spread-spectrum radio protocol for GCS-to-drone command link
M_b	Battery mass (kg) in the endurance optimality theorem
M_f	Fixed subsystem mass (kg) — everything except battery
MTOW	Maximum Take-Off Weight — total mass including battery and payload
PDB	Power Distribution Board — custom PCB distributing battery power to ESCs and avionics
RPN	Risk Priority Number — FMEA severity $\times$ occurrence $\times$ detectability (1–1,000)
RTH	Return to Home — autonomous flight back to launch point triggered by low SOC or link loss
SIL	Software-in-the-Loop — simulation with all software running on PC, no physical hardware
SOC	State of Charge — battery charge level as percentage of full capacity
SPF	See, Point, Fly — visual-language UAV navigation framework achieving 92.7% indoor success
SSR	Solid-State Relay — electronic switch gating dock pogo-pin power until R-match confirmed
TDMA	Time Division Multiple Access — scheduled radio channel access protocol for fleet mesh
THD	Total Harmonic Distortion — ratio of harmonic content to fundamental in motor current
TRL	Technology Readiness Level — NASA/ESA 1–9 scale for technology maturity

Term	Full Form / Definition
UWB	Ultra-Wideband — short-range (100 m) precision ranging radio; DWM3000 used for federation EKF
VIO	Visual-Inertial Odometry — camera + IMU fusion for position estimation without GPS
$W^{1.5}$	The hover power scaling law: $P_{\text{hover}} \propto (\text{total mass})^{1.5}$ from actuator disc theory

Table 5. Technical glossary for all terms used across the document set.

9. PROJECT MILESTONES AND DECISION LOG

Date	Milestone / Decision	Evidence / Trigger	Impact
Jan 2026	Architecture pivot: abandon per-drone optimisation; adopt fleet model	$W^{1.5}$ analysis showed $M_{b_opt} = 2 \times M_f$ impossible without removing compute	Entire project reframed; fleet logistics becomes core contribution
Feb 2026	FOC adopted as primary ESC strategy	13 dB SPL + 11.3× vibration reduction measured in anechoic chamber	Acoustic and IMU quality improvements cascade through to SOC accuracy
Mar 2026	Isolated power rail (PM02D) made mandatory	12% brownout rate discovered under full AI load; PM02D eliminates it	Power architecture locked; this is non-negotiable for any AI-carrying drone
Mar 2026	Carrier drone identified as engineering void	Fleet endurance model requires carrier but no design existed	K1/K2 specification work initiated (DOC4)
Apr 2026	IR beacon docking concept proposed	Analysis showed AprilTag guidance insufficient (4.8 Hz) for 8 m/s wind	IRDG-01 PCB design initiated; IP provisional filing planned
Apr 2026	Federated EKF proposed for GPS-denied nav	GPS-denied architecture inconsistency found between paper versions	DOC3 section written; SIL test S-04 planned
May 2026	Document set created (DOC0–DOC8)	Project at 70-80% completeness; formalised for publication and IP	First complete documentation of all project knowledge
TBD	W1 stripped worker first flight	Hardware built after W1 BOM finalised	First validation of 159 min endurance prediction
TBD	K1 first docking cycle	K1 fabricated; F4 test completed	First validation of fleet rotation logistics
TBD	Full fleet 2-hour demonstration	F7 test completed	Primary project milestone; enables journal submission update

Table 6. Project milestone and decision log.